INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

SALTLAKE

(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – II, Second Semester Examination, March - 2025

Subject Code: BCACC202

Subject Name: Object Oriented Programming with C++

Full Marks: 30 | Set No: 1 | Duration: 1 Hour

Part - A
Answer any 5 out of 6 questions
Each question carries 2 marks (2 × 5)

1. Define inheritance. 2

2. What is a pointer? How is it different from a reference variable? 2

3. Explain the concept of function overriding. 2

4. How does polymorphism relate to inheritance? 2

5. What are virtual functions? 2

6. Describe the use of throw and catch in exception handling. 2

Part - B
Answer any 2 out of 3 questions
Each question carries 5 marks (5 × 2)

7. Demonstrate how seekg() and seekp() manipulate file pointers with an example. 5

8. Explain the use of 'this' pointer in C++ with an example. 5

9. How does dynamic memory allocation work with pointers in C++? Provide an example. 5

Part - C
Answer any 1 out of 2 questions
Each question carries 10 marks (10 × 1)

10. Explain multiple and multilevel inheritance in C++ with proper examples. 10

11. Write a program demonstrating the use of virtual base classes in C++. 10